



CC 141

Discrete Structures

Department of Computer Science
School of Systems and Technology

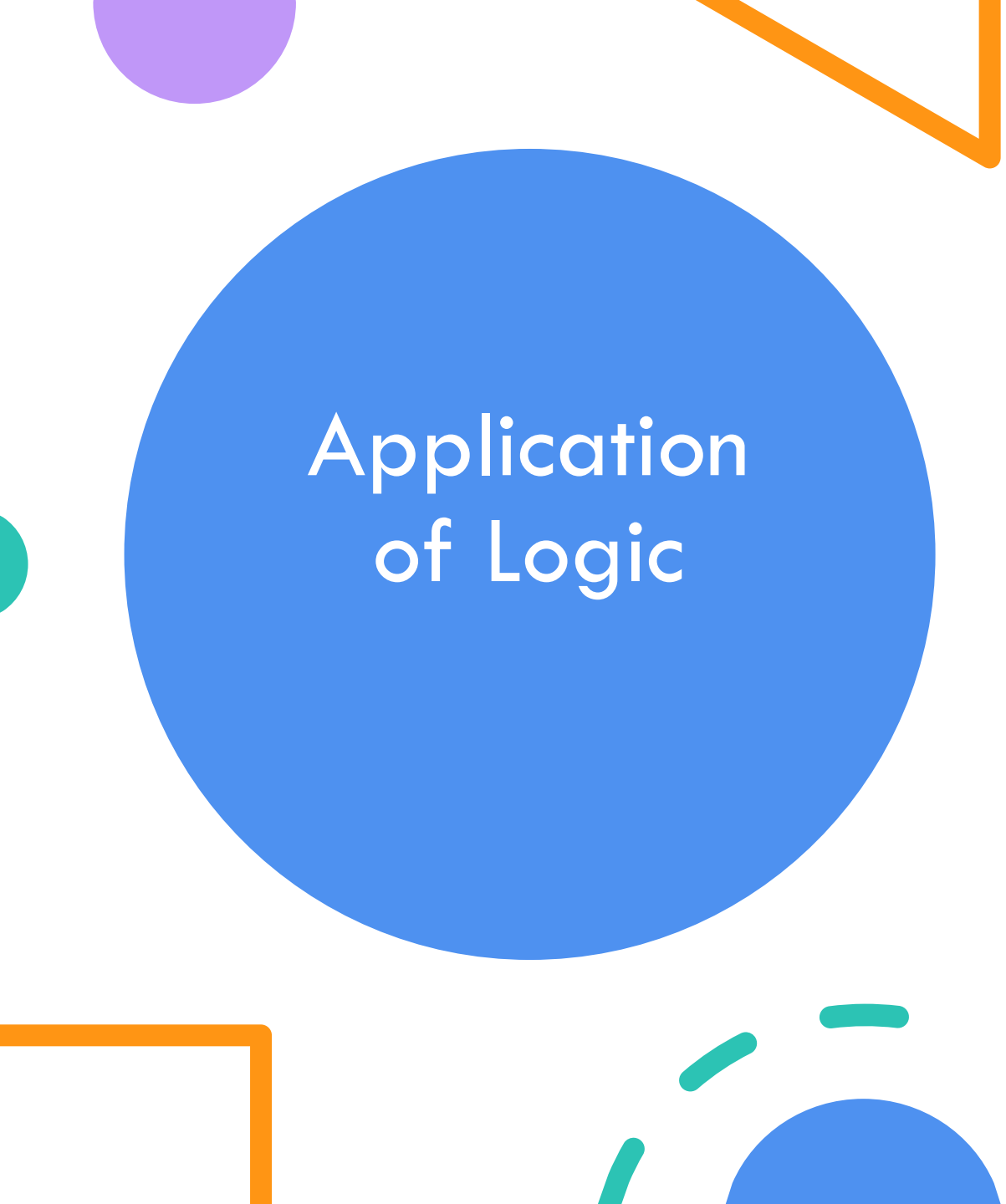
University of Management and Technology
Lahore, Pakistan

A decorative green dashed line consisting of several short, slightly curved segments, arranged in an arc on the left side of the blue circle.

Section 1.2

Application of Propositional Logic

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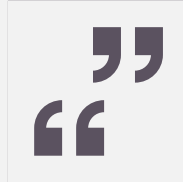
Application of Logic

- Translating English Sentences
- System Specifications
- Boolean Searches
- Logic Puzzles
- Logic Circuits

Translating English Sentences



Step 1 : find logical connectives



Step 2: break the sentence into basic propositions



Step 3: rewrite the sentence in propositional logic

Translating English Sentences

Example:

- If you are senior citizen **and** it is a Tuesday, **then** you can have free coffee

Step 1 : find logical connectives

Step 2: break the sentence into basic propositions

Translating English Sentences

Step 2: break the sentence into basic propositions

Example:

if you are senior citizen and it is a Tuesday then

a

b

you can have free coffee

c

Translating English Sentences

Step 3: rewrite the sentence in propositional logic

Example:

if you are senior citizen and it is a Tuesday then

a

b

you can have free coffee

c

$$a \wedge b \rightarrow c$$

System Specifications

- "The automated reply cannot be sent when the file system is full"



System Specifications

p ="The automated reply can be sent"

q ="The file system is full."

Then $\neg p$ represents "It is not the case that the automated reply can be sent," which can also be expressed as "The automated reply cannot be sent."



System Specifications

"The automated reply cannot be sent when the file system is full"

- p = "The automated reply can be sent"
- q = "The file system is full."

- Consequently, our specification can be represented by the conditional statement
 - $q \rightarrow \neg p$

Boolean Searches





Boolean Searches

Logical connectives are used extensively in searches of large collections of information, such as indexes of Web pages. Because these searches employ techniques from propositional logic, Links they are called Boolean searches

Example: Web Search:

UMT AND CS AND DISCRETE

UMT OR CS OR DISCRETE

The left side of the slide is decorated with various geometric shapes: a purple circle at the top left, a teal vertical line below it, a large blue semi-circle in the middle left, a teal vertical line to its left, an orange triangle at the top right, a purple circle in the middle right, a large blue circle at the bottom center, a teal dashed line above it, and an orange square at the bottom left.

Logical Puzzles

Puzzles that can be solved using logical reasoning are known as logic puzzles

Examples – Logical Puzzles



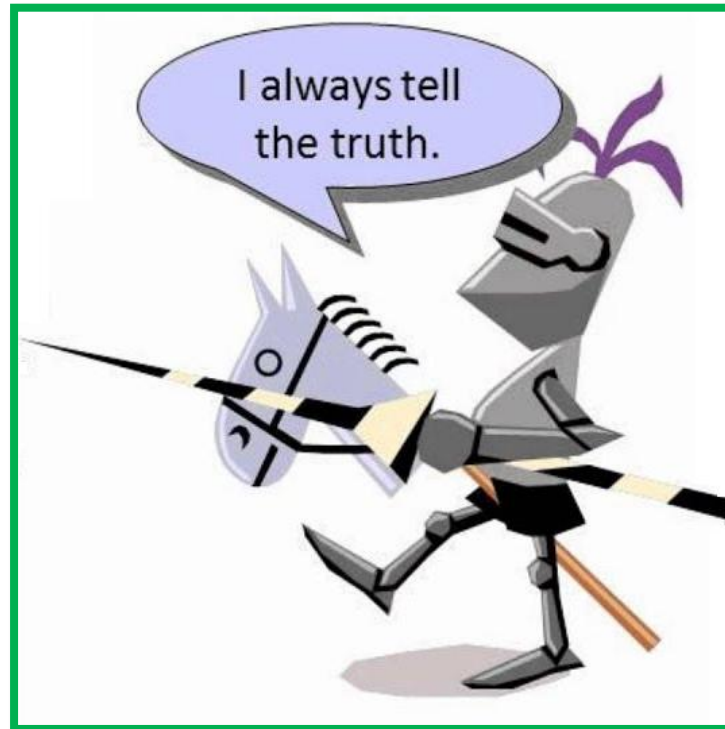
Examples – Logical Puzzles

EXAMPLE

An island has two kinds of inhabitants, knights, who always tell the truth, and their opposites, knaves, who always lie.

You encounter two people *A* and *B*.

What are *A* and *B* if *A* says “*B* is a knight” and *B* says “The two of us are opposite types?”



RAYMOND SMULLYAN

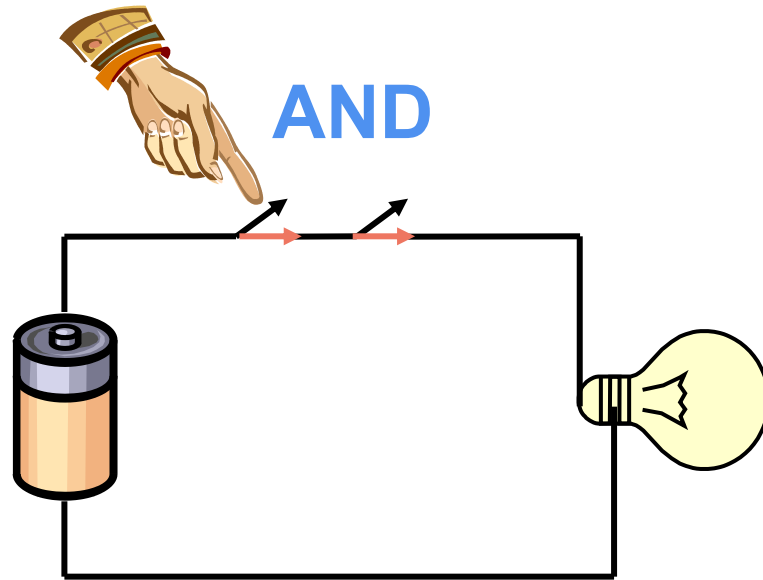
The slide features several decorative geometric shapes on the left side. At the top left is a purple circle. Below it are two vertical teal dashes. To the right of these is a large blue semi-circle. Further right is a purple circle outline. At the bottom left is an orange square outline. Above it are two teal dashes. At the bottom center is a large blue circle. Above it are three teal dashes. At the top right is an orange triangle outline.

Logical Circuits

Switches “in series”

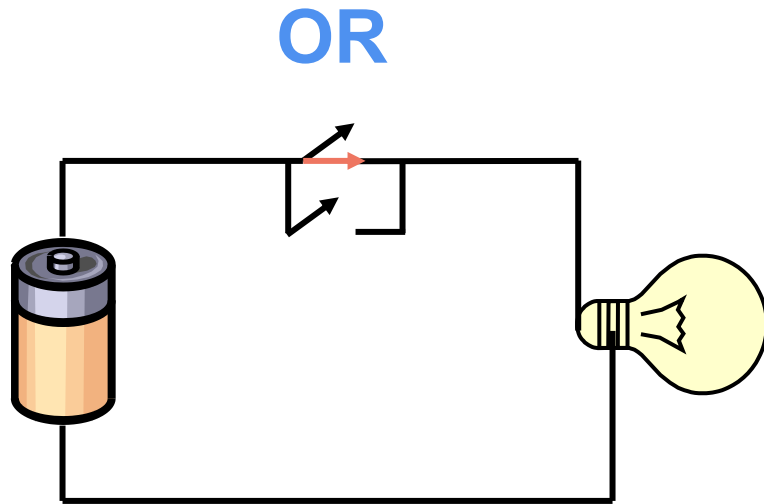
Switches “in parallel”

Switching Circuits – Series Circuit



Switches		Light Bulb
P	Q	State
Closed	Closed	On
Closed	Open	Off
Open	Closed	Off
Open	Open	Off

Switching Circuits – Parallel Circuit



Switches		Light Bulb
P	Q	State
Closed	Closed	On
Closed	Open	On
Open	Closed	On
Open	Open	Off

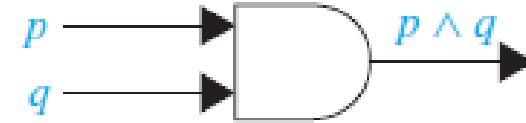
Logic Circuits



Inverter



OR gate

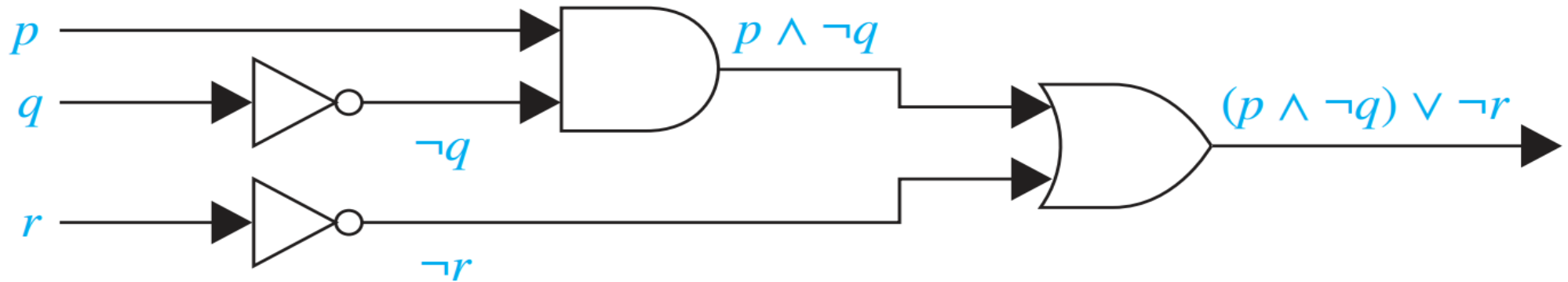


AND gate

FIGURE 1 Basic logic gates.

Logic Circuits

- Determine the output for the combinational circuit in following Figure

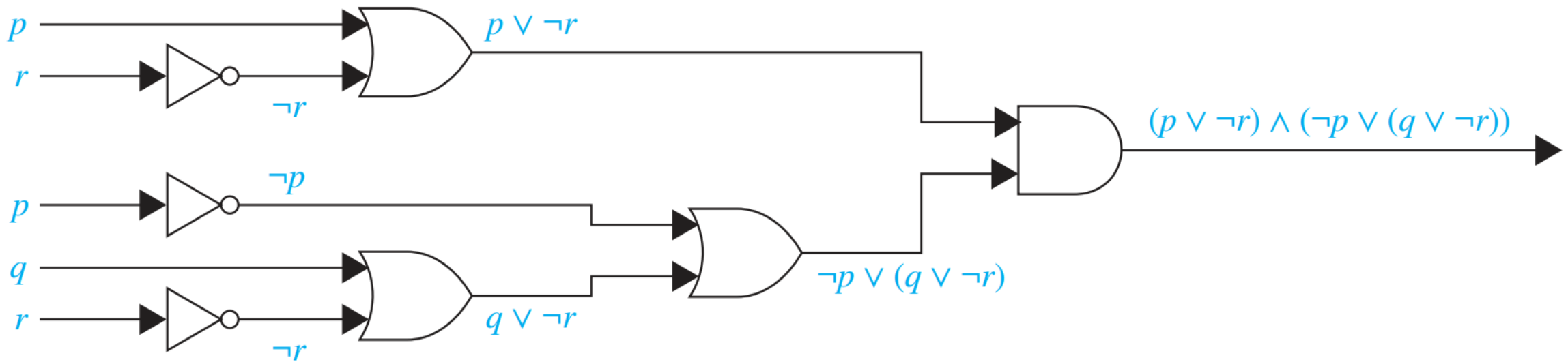


Logic Circuits

- Build a digital circuit that produces the output

$$(p \vee \neg r) \wedge (\neg p \vee (q \vee \neg r))$$

- when given input bits p , q , and r .



Summary

1.2 Application of Propositional Logic

- ✓ Translating English Sentences
- ✓ System Specifications
- ✓ Boolean Searches
- ✓ Logic Puzzles
- ✓ Logic Circuits

